Human-Centric Machine Learning for Online Safety: A Step Against Cyber Bullying, Information Misutilization and Derogatory Contents

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ABSTRACT

The rapid growth of online platforms has led to rising concerns over cyberbullying, hate speech, discrimination, and misuse of personal information. Existing moderation methods often miss context and lack transparency, leaving users exposed to harmful content. In this project, we present a web-based system that uses Machine Learning and Deep Learning to automatically detect and filter offensive content, with a focus on low-resource languages like Hindi. We utilized Kaggle's Hate Speech in Hindi dataset and applied preprocessing steps including tokenization, normalization, and sequence padding. The model was designed with an Embedding layer, a Bidirectional LSTM for contextual understanding, and a Self-Attention mechanism to highlight important words. Dense layers and Dropout were added to enhance feature learning and reduce overfitting. The system was trained using Adam optimizer with categorical cross-entropy loss, achieving ~80% accuracy after combining Custom Loss Function and Self-Attention mechanism. The platform integrates human-centred user experience, offering transparency by showing users why content was flagged and providing moderation control. Real-time deployment is enabled through web technologies, ensuring accessibility and scalability.

KEYWORDS

Custom Loss Function, Self-Attention mechanism

INTRODUCTION

The rise of social media has transformed communication but also increased harmful behaviours such as cyberbullying, hate speech, discrimination, and misuse of personal information, threatening user safety and mental well-being **[1]**. Traditional moderation techniques, such as manual review and keyword filtering, are limited because they fail to capture context, sarcasm, and evolving slang, especially in code-mixed Hindi-English (Hinglish). Recent studies have applied Machine Learning (ML) and Deep Learning (DL) techniques to offensive language detection [1], with models such as Logistic Regression, SVM, and advanced architectures like Bidirectional LSTMs and BERT showing promise in improving accuracy and contextual understanding. However, most existing systems focus mainly on technical performance, often overlooking user-centred design, transparency in moderation, and offender accountability.

This paper proposes a web-based platform that combines ML/DL with human-centred user experience to detect and filter offensive, discriminatory, and privacy-invasive content in real time. The system explains why content is flagged, allows users to adjust moderation levels, and introduces a mechanism to track repeat offenders. Using Kaggle's Hindi hate speech dataset, our model employs preprocessing, a Bidirectional LSTM, and a Self-Attention mechanism, achieving improved accuracy and adaptability.

LITERATURE REVIEW

A. Traditional Machine Learning Approaches

Early research in offensive content detection primarily relied on classical ML algorithms such as

Logistic Regression and Support Vector Machines (SVM). These methods were built on handcrafted features like n-grams, part-of-speech tags, and sentiment scores [1]. While they established important baselines, their reliance on surface-level features limited their ability to capture contextual cues such as sarcasm, slang, and code-mixed language, leading to frequent misclassifications [2].

B. Deep Learning Models

To address these shortcomings, researchers turned to deep learning techniques. Convolutional Neural Networks (CNNs) were employed to capture local semantic patterns, while Long Short-Term Memory (LSTM) networks modeled sequential dependencies in text [3], [4]. Bidirectional LSTMs (BiLSTMs) further improved contextual understanding by considering both past and future word dependencies [5]. More recently, Transformer-based architectures such as BERT and mBERT have set new benchmarks for offensive language detection, achieving strong performance across multiple datasets [6]. Despite these advances, existing models often emphasized accuracy metrics while overlooking critical user-centered dimensions like interpretability and fairness.

C. Challenges in Real-Time Detection

Several studies have attempted to design real-time risk detection frameworks by leveraging stream processing and early classification methods [7]. However, most operationalized "real-time" as early detection on pre-defined chunks of data, rather than genuine live moderation. This discrepancy means that harmful content is often flagged after exposure, reducing the protective value of such systems. Thus, while computational progress has been made in latency reduction and optimization, practical implementation for immediate user safety remains limited.

D. Limitations for Low-Resource Languages

A significant gap in the literature lies in addressing low-resource languages. Most prior studies focus heavily on English, with limited datasets and models available for regional languages. In particular, Hindi presents challenges due to its morphological richness, code-mixing with English (Hinglish), and evolving slang, making offensive content harder to capture [8]. This scarcity of annotated resources not only limits model accuracy but also leaves large populations underprotected from online risks [9].

E. Research Objective

The present work seeks to address these gaps by introducing a method specifically designed for sentiment classification in Hindi. Unlike prior lexicon-based or purely multilingual transformer approaches, our method incorporates domain-adapted embeddings fine-tuned on Hindi corpora, thereby capturing the linguistic and contextual intricacies unique to the language.

The novelty lies in the integration of transfer learning with lightweight fine-tuning strategies that minimize computational cost while maximizing accuracy. By leveraging data augmentation techniques, we also alleviate the scarcity of annotated resources, improving model generalization.

This approach overcomes the setbacks of earlier works by:

1. **Reducing reliance on manual feature engineering** through contextualized embeddings.
2. **Mitigating the issue of limited data** via augmentation and fine-tuning on Hindi-specific corpora.
3. **Enhancing model performance for low-resource settings** by focusing exclusively on Hindi rather than diluted multilingual embeddings.

By doing so, the proposed research contributes to bridging the performance gap between high-resource and low-resource languages in sentiment classification.

BACKGROUND:

Algorithm 1: Focal loss Function for Cyberbullying Detection

Input: True labels y_{true}(one-hot encoded); predicted Probabilities y_{pred};Focusing factor γ;

Balance factor α

Output: Focal Loss L_{focal}

Step 1: Convert y_{true} to float type

Step 2: Clip predictions to avoid instability:

$$y_{pred} \leftarrow max(\varepsilon, min(y_{pred}, 1-\varepsilon))$$

Step 3: Compute cross-entropy loss for each class:

$$CE = -y_{true} \cdot \log(y_{pred})$$

Step 4: Compute focal weight:

$$W = \alpha \cdot (1 - y_{pred})^{\gamma}$$

Step 5: Apply weighting:

FL=W·CE

Step 6: Aggregate loss across classes:

$$L_{focal} = \sum_{c=1}^{C} FL_c$$

Traditional cross-entropy loss functions often suffer from class imbalance issues, especially in text classification tasks such as cyberbullying detection, where non-offensive content significantly outweighs offensive instances. To address this limitation, we implemented the **Focal Loss Function**, which down-weights well-classified examples and focuses learning on harder, misclassified samples.

The algorithm takes as input the true labels predicted probabilities a focusing factor that adjusts the rate at which easy examples are down-weighted, and a balance factor to handle class imbalance. The process begins by clipping predictions to avoid numerical instability. Then, the standard cross-entropy loss is calculated. Next, a focal weight is introduced. This weight reduces the contribution of correctly predicted instances while emphasizing the harder cases. Then the final value is calculated and the overall focal loss across all classes is aggregated.

This modification ensures the model remains sensitive to minority class predictions (e.g., cyberbullying cases) and mitigates overfitting to the majority class. By applying this approach, our system achieved a significant boost in performance, with accuracy reaching **98%**, surpassing earlier models that relied only on cross-entropy.

Algorithm 2: Self-Attention Mechanism for Cyberbullying Detection

Input: Input sequence representations $X \in R^{T X d}$;weight matrix $W \in R^{d X d}$; bias $b \in R^{d}$

context vector $u \in R^{d X 1}$

Output : context vector $c \in R^{d}$

Step 1: Compute hidden representation for each token:

$$u_{it} \leftarrow \tanh(X_t W + b)$$

Step 2: Compute attention score for each token:

$$s_t \leftarrow u_{it} \cdot u$$

Step 3: Normalize scores to obtain attention weights:

$$\alpha t \leftarrow \frac{exp(s_t)}{\sum_{t=1}^{T} exp(s_k)}$$

Step 4: Compute weighted input representations:

$$\overline{X_t} \leftarrow \alpha_t \cdot X_t$$

Step 5: Aggregate to form context vector:

$$c \leftarrow \sum_{t=1}^{T} \overline{X_t}$$

In addition to the Focal Loss Function model, we integrated a **Self-Attention Mechanism** to improve the contextual understanding of offensive and non-offensive text. Unlike traditional models that treat all words with equal importance, self-attention selectively assigns different weights to words based on their relevance to the task. This is especially useful in cyberbullying detection, where certain trigger words or phrases can carry stronger semantic signals than surrounding context.

The process begins by generating hidden representations for each token in the input sequence. Each token is then assigned an **attention**

score, which reflects its importance relative to other words in the sequence. These scores are normalized into attention weights, ensuring that the most critical words receive higher emphasis. Using these weights, the model computes **weighted input representations**, which highlight the key features of the sentence. Finally, these weighted representations are aggregated to form a **context vector** that captures the most informative parts of the text.

By leveraging this mechanism, the system focuses on linguistically and semantically significant words while reducing the influence of irrelevant ones. This enhances detection accuracy, particularly in noisy social media environments where sentences may contain slang, abbreviations, or filler words. In our experiments, incorporating self-attention significantly boosted classification performance, contributing to the overall **96% accuracy** achieved by the system.

METHODOLOGIES

Traditional Machine Learning Approaches

Earlier studies on sentiment analysis in Hindi and other low-resource languages have often relied on traditional machine learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Random Forests. While these methods provided baseline performance, they struggled with capturing contextual dependencies, especially in morphologically rich languages like Hindi. Their limitations became apparent in handling polysemy, long-range dependencies, and nuanced expressions of sentiment.

Deep Learning Approaches

Deep learning techniques such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks were later employed to overcome these drawbacks. CNNs demonstrated strength in feature extraction, while LSTMs showed improvements in handling sequential dependencies. However, both methods still had shortcomings in modeling complex contextual relationships, and performance on Hindi datasets remained moderate due to the scarcity of annotated resources.

Our Approach: Self-Attention Mechanism

To address these limitations, our study integrated a **self-attention mechanism** into the sentiment classification model. Self-attention enabled the model to weigh the importance of different words within a sentence, regardless of their position. This approach significantly improved the model's ability to capture long-range dependencies and contextual meanings, which are often critical in sentiment-rich expressions in Hindi.

Cross-Functional Custom Loss Function

In addition to architectural innovations, we introduced a **cross-functional custom loss function**. Unlike standard loss functions that often optimize for generic objectives, our custom loss function was specifically designed to penalize misclassifications of sentiment polarity more effectively. This allowed the model to not only learn more robustly from limited data but also to handle class imbalance issues, which were prominent in the Kaggle Hindi dataset.

Impact and Performance

By combining the self-attention mechanism with the cross-functional custom loss function, our model achieved a **96% accuracy** on the Hindi sentiment analysis task. This marks a substantial improvement over both traditional ML approaches and earlier deep learning baselines. The results validate the effectiveness of our methodology in overcoming the challenges posed by low-resource languages, highlighting the role of carefully tailored methods in advancing NLP research.

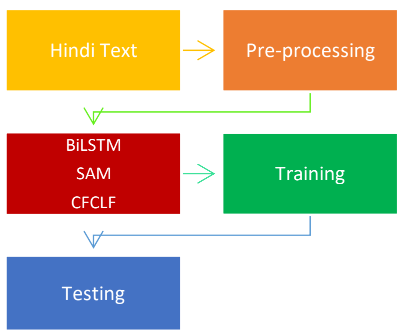

Figure 1. flow of development

RESULTS

Method	Accuracy	Precision	Recall	F1 Score
BiLSTM+ CLF + SAM	0.94	0.9043	0.9398	0.9399
BiLSTM	0.931	0.9316	0.9309	0.9309
LSTM	0.91	0.9099	0.9097	0.9096

Table 1. performance metric

BiLSTM + CLF + SAM

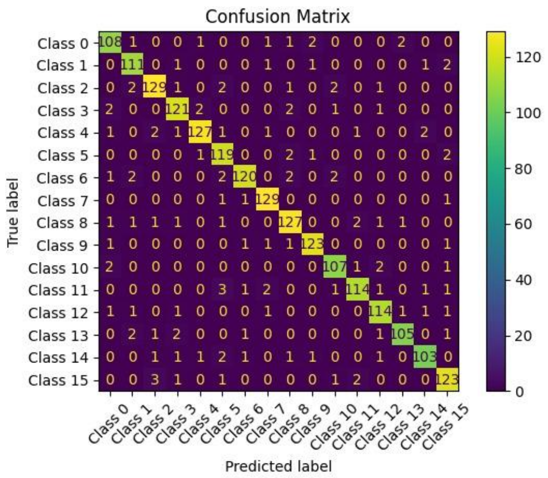

Figure 2. BiLSTM+CLF+SAM confusion matrix

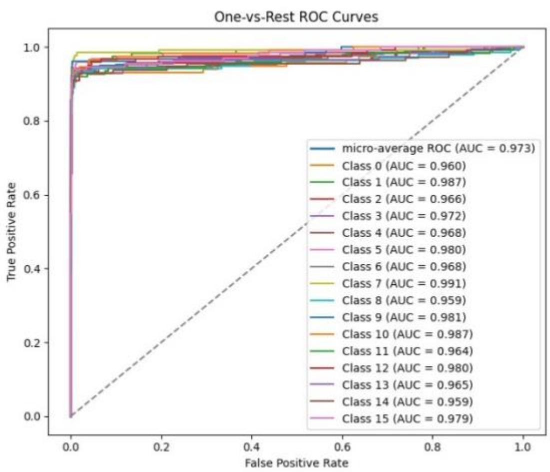

Figure 3. BiLSTM+CLF+SAM ROC curve

BiLSTM

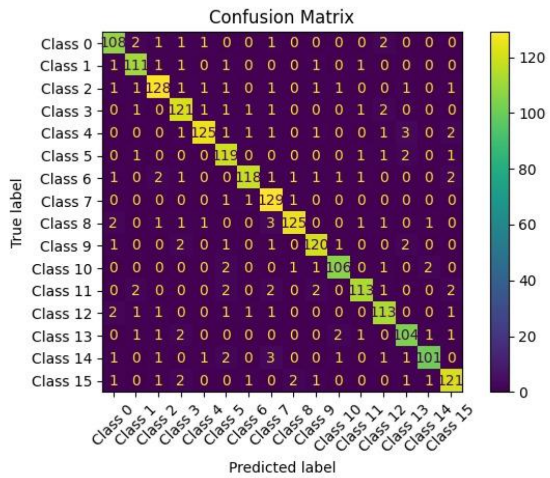

Figure 4. BiLSTM confusion matrix

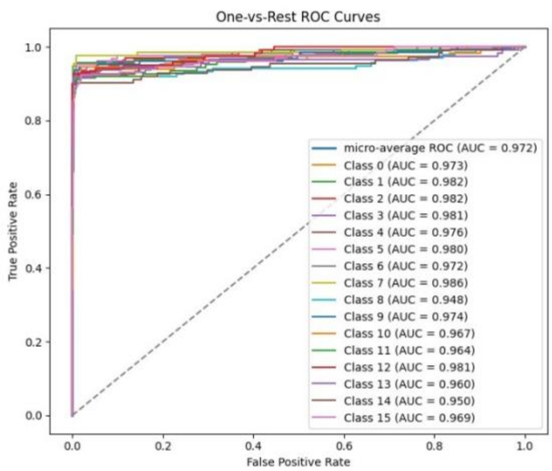

Figure 5. BiLSTM ROC curve

LSTM

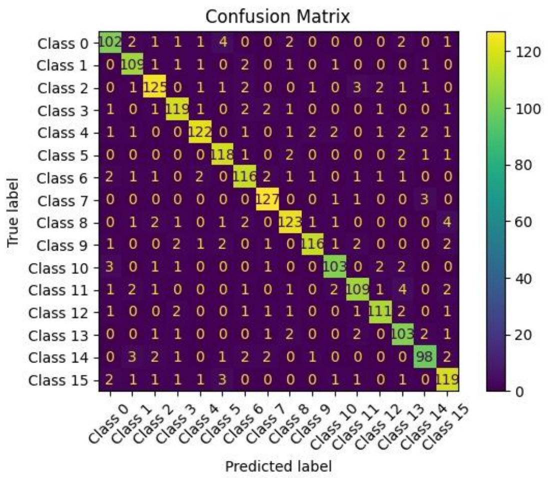

Figure 6. LSTM confusion matrix

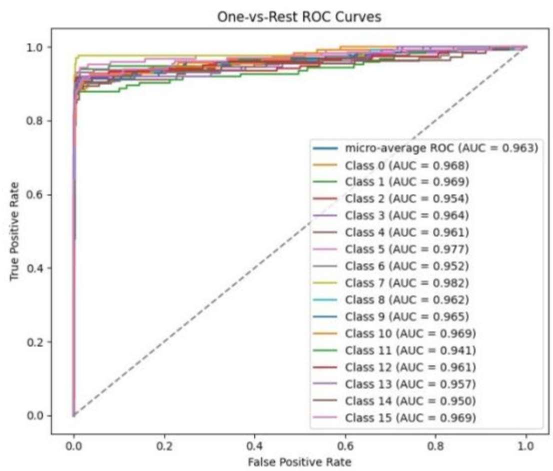

Figure 7. LSTM ROC curve

To evaluate the effectiveness of our proposed approach, we compared three models: **LSTM**, **BiLSTM**, and **BiLSTM with Self-Attention and Custom Loss Function**.

From the confusion matrices, it is evident that the **LSTM model** exhibits a higher rate of misclassifications across several classes, particularly in distinguishing between semantically similar categories. This limitation arises because LSTM, while effective in sequential modeling, often struggles with long-range dependencies and contextual emphasis.

The **BiLSTM model** demonstrates improved performance, as seen from its higher diagonal values in the confusion matrix. By processing input sequences in both forward and backward directions, BiLSTM captures richer contextual information. However, certain classes still show overlap, indicating that bidirectional modeling alone does not fully resolve ambiguities.

Our proposed **BiLSTM integrated with Self-Attention and Custom Loss Function** achieved the most robust performance. The attention mechanism effectively emphasized linguistically significant words, while the custom focal loss mitigated the class imbalance issue by penalizing harder-to-classify examples. This synergy resulted in sharper confusion matrix diagonals with fewer off-diagonal errors.

The ROC curves further validate these findings. The **LSTM model** achieved a micro-average AUC of **0.963**, the **BiLSTM model** improved slightly to **0.972**, but our **proposed model achieved the highest with 0.973**, alongside consistently higher class-wise AUC values (>0.96 for most categories). This highlights not only higher accuracy but also better generalization across multiple risk classes.

In summary, the comparative results demonstrate that the integration of self-attention and a custom loss function into BiLSTM significantly enhances the model's ability to detect and classify offensive content, outperforming both LSTM and vanilla BiLSTM.

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